

# Opening doors to data

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Canadian Research Data Centre Network

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# Introduction

- Thanks to SSHRC for supporting this “Opening doors to data” project via the Connections Grant program: RDM stream
- Thanks to CRDCN and its funders: CFI, CIHR, SSHRC, Provincial Partners and Collaborating Institutions.

Please access the project folder for this workshop:

[www.github.com/CRDCN/RAD\\_training](https://www.github.com/CRDCN/RAD_training); or

<https://doi.org/10.17605/OSF.IO/ATWNJ>

# Introduction

- Evolution of data discovery & open science
- Journal pressures
- Council pressures
- University pressures

# Containing this

- Today we're talking about restricted access data
- Not all sensitive data, and not open data, but data that have an access process or mechanism

# Targets

- FAIR data
- “As open as possible, as closed as necessary”
- This is not a binary, but a continuum

# Researcher support for discoverability

F - *Findable*

# Researcher support for discoverability

## F - *Findable*

Can anyone discover that your data even exist?

Can others even figure out where the data you used are?

Concepts:

- Persistent Identifiers
- Indexing (including metadata)

# Researcher support for discoverability

*A - Accessible*



# Researcher support for discoverability

## *A - Accessible*

Can others access the data you used?

Can they figure out HOW to do so?

Concepts:

- Data accessibility statement
- Access metadata
- Transparent process

# Researcher support for discoverability

I - *Interoperable*

# Researcher support for discoverability

## I - *Interoperable*

### Concepts:

- Open source
- Machine-readable
- Metadata
- Control vocabularies

# Researcher support for discoverability

R - *Reusable*

# Researcher support for discoverability

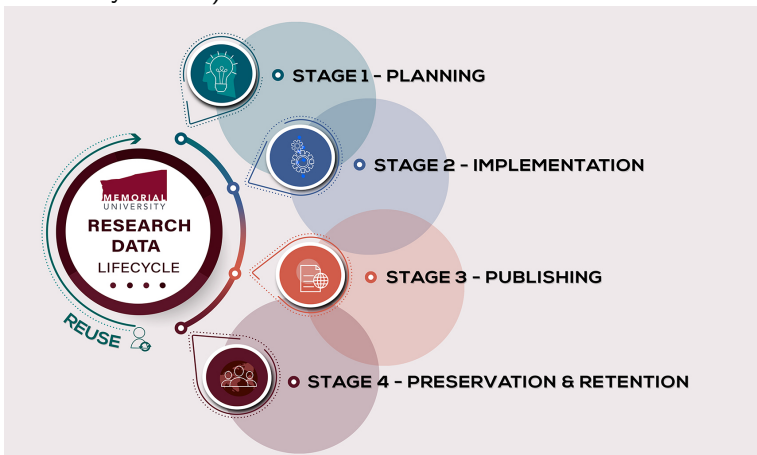
R - *Reusable*

Concepts:

- Provenance
- Licensing
- Archiving

# Data lifecycle

- Image of data lifecycle & description (credit: Memorial University OVPR)



## Case study: CBS

- Canadian Blood Services offers secondary use research data (that is, databases about donors that you can request)
- Suppose you were a researcher interested in research on data about donors

Based on information you're able to get online:

- 1 Think about a research question and evaluate whether you could answer it
  - 2 Outline the process you would follow to get the data
- We'll take up these questions and discuss how "FAIR" we think the data are.

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# Takeup (GA)

# FAIR Restricted data

- Findability doesn't **need** to be affected, but often is
- Accessibility might mean something different
- Interoperability can sometimes be tricky
- Reusability can be better, but this requires effort

# Findability

- Organizations providing data as a non-priority activity
- Resources to make it findable may not even be considered
- Knowledge of how to make data discoverable might not exist
- Even where a core dataset from an academic source exists there can be
  - A research team primarily using the data and making it available is secondary (see item 1)
  - Because it isn't open, it's not posted, and making it findable is a separate activity (where with open data it's often put into a service that manages both curation and discoverability)

# Accessibility

- Is it even considered?
  - I would argue yes, very seriously, but not with a lens of FAIR
  - Some of this is foundational see *Read et al. 2024*

# Interoperability

- This goes hand-in-hand with findability and suffers the same resourcing issue
- Metadata can have the potential to disclose individuals
- Restricted datasets *don't generally have good metadata*

# Reusability

- Data often belong to an organization and so don't suffer the same risks that data held by individuals do
- Similar to Interoperability/Findability issues though, improper data management makes data less reusable

# What would success look like? (GA)

# Data management planning

- *DMP template tool*
  - Consider what anyone following on from you will be starting from. Everything that got you from that step to another point becomes part of your research data
  - Not every part of the project data will therefore be restricted and you need to plan for that



# Data Accessibility Statements

- “Access available on request” NOT sufficient
- Encourage data source to template this language! If no, DIY with review

# Data Accessibility Statement Contents

# Data Accessibility Statement Contents

- Who can access
  - Rank?
  - Research themes?
  - Citizenship?
- Terms of access
  - Consultation?
  - Ethics?
  - Proposal?
  - Citation?
  - TRE?
- Licensing/Agreements
  - Ex. CC-BY vs. CC-BY-NC
- Costs

# Data Accessibility Statements

- Thorough description of the access process, eligibility requirements links to info, financing considerations, and licensing info/terms-of-use

## Data discovery efforts

- Any datasource can now be indexed in Lunarix relatively easily.
- Metadata only deposits where there is some info to provide to potential users.
- Metadata only deposits also create a PID (persistent identifier) which makes the data a lot easier to find for someone who wants to use it in the future because *YOU CAN CITE IT*

## Preservation efforts

- If a datasource overwrites my data with a new copy and doesn't say anything then someone trying to recreate my work will be very confused at best
- Ideally new versions get new PIDs and older versions point to the newest, or everything is gathered under one PID with versioning.
- Frequency of versioning will depend on frequency of access and preferences of organizations.

# How do I talk to my data source?

- Raise the issue
- Highlight the benefits
- Point to resources

## Case study; indexing in Lunaris

- Lunaris is a service of the Digital Research Alliance of Canada
- It collates metadata from a variety of sources
- Data sources can work with Lunaris to have their datasets indexed



# Lunaris page

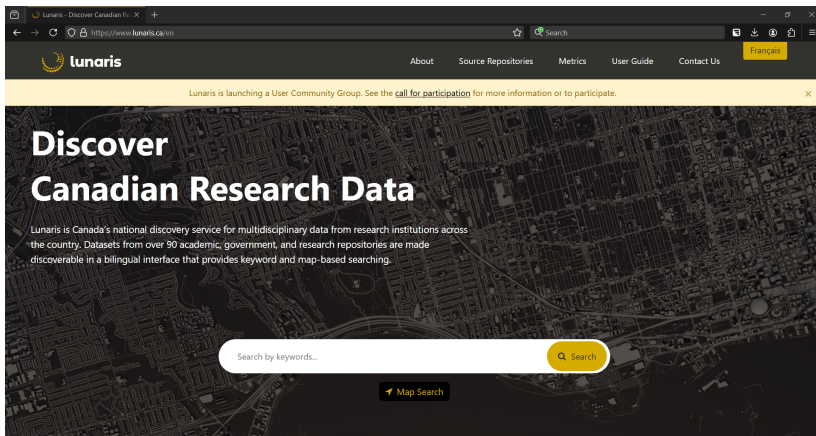


Figure 2: Lunaris webpage screenshot

# System needs

- Lunaris has a metadata schema (list of information it can process and display)
- Any data source can provide a database of this information about their datasets to be “harvested”
- Work with Lunaris team to determine how the data should be formatted (samples from our case study in the workshop folder)

# Case study

The screenshot shows the CRDCN website with the 'Publications' tab selected. The main heading is 'Farm Management Survey' for the years 2017 - 2021. A sidebar on the right lists subjects: Agriculture and food, Environment, Environmental protection, and Land use. Below the heading, there are tabs for Overview, Related Publications, and Related Data. The 'Overview' tab is active, showing a summary of the survey and a table of available cycles. The summary text describes the survey as a collaborative project between Statistics Canada and Agriculture and Agri-Food Canada, conducted every five years. The table lists the 2017/2021 cycle as the 'Farm Management Survey'. On the right, there is a 'Documentation' section with a 'STATSCAN' link and a 'Publication Note' section with a 'See a sample' button.

Canadian Research Data Centre Network

Home About VRDC Programs Initiatives News Data Publications Français

Home > Data > Datasets > Farm Management Survey

## Farm Management Survey

2017 - 2021

Subjects

- Agriculture and food
- Environment
- Environmental protection
- Land use

Overview Related Publications Related Data

### Overview

#### Summary

The Farm Management Survey (FMS), conducted every five years, is a collaborative project between Statistics Canada and Agriculture and Agri-Food Canada. The FMS contributes to Agriculture and Agri-Food Canada's work on measuring selected management practices in the agriculture sector. The information generated from this survey informs federal and provincial policy decisions in the sector. FMS data were collected using a sample of 18,000 farms selected to be representative of 51% of Canadian production of dairy cows, beef, poultry, pork, field crops, forage crops, and fruit, vegetable, berry and nut crops. Small farms were automatically excluded, and large ones were considered "must take" because of their national influence on farm practices. The survey aims to produce estimates for seven farm types at both the provincial and ecological region levels. Each respondent filled out the questionnaire for only one of these commodities based on the significance of their production nationally and within their region. Data from the Census of Agriculture were used to identify the importance of each farm's production.

#### Available Cycles

| Year      | Name                   |
|-----------|------------------------|
| 2017/2021 | Farm Management Survey |

#### Documentation

STATSCAN

#### Publication Note

All publications (e.g. scientific articles, reports, dissertations, theses) and presentations based on a dataset available in the RDCs should include an acknowledgment of the support provided by granting councils (SSHRC, CIHR, CFI), Statistics Canada and host university.

See a sample →

Figure 3: CRDCN data webpage screenshot

# Case study

The screenshot shows the Canadian Research Data Centre Network (CRDCN) website. The header includes the CRDCN logo and navigation links: Home, About, VRDC, Programs, Initiatives, News, Data, Publications, and Français. The main content area is titled "Farm Management Survey" with the years "2017 - 2021". A "Subjects" sidebar lists "Agriculture and food", "Environment", "Environmental protection", and "Land use". Below the title, there are tabs for "Overview", "Related Publications", and "Related Data". The "Overview" tab is active, showing a "Summary" section. The summary text describes the Farm Management Survey (FMS) as a collaborative project between Statistics Canada and Agriculture and Agri-Food Canada, conducted every five years. It mentions that FMS data were collected using a sample of 18,000 farms selected to be representative of 51% of Canadian production of dairy cows, beef, poultry, pork, field crops, forage crops, and fruit, vegetable, berry and nut crops. Small farms were automatically excluded, and large ones were considered "must take" because of their national influence on farm practices. The survey aims to produce estimates for seven farm types at both the provincial and ecological region levels. Each respondent filled out the questionnaire for only one of these commodities based on the significance of their production nationally and within their region. Data from the Census of Agriculture were used to identify the importance of each farm's production.

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See a sample ->

Figure 4: CRDCN webpage data screenshot

# Case study

|    | A            | B                                                                                                                                                             | C | D | E | F | G | H | I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | J | K                  | L                    | M                             | N               | O        | P | Q    | R    |                |
|----|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------------------|----------------------|-------------------------------|-----------------|----------|---|------|------|----------------|
|    | Formal title | Dataset / Analysis / Data Type / Release / Platform / Dictionary                                                                                              |   |   |   |   |   |   | Summary / Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   | Source / URL / Ref | URL / Ref / Subjects | Access / Rights               | Access / Rights | Comments |   |      | Date |                |
| 18 | 20003        | Intercrop Farm Management Survey (FMS) conducted every five years, is a collaborative project between Statistics Canada and Agriculture and Agri-Food Canada. |   |   |   |   |   |   | The Farm Management Survey (FMS) conducted every five years, is a collaborative project between Statistics Canada and Agriculture and Agri-Food Canada.<br><br>The FMS contributes to Agriculture and Agri-Food Canada's work on measuring selected management practices in the agriculture sector. The information generated from this survey informs national and provincial policy decisions in the sector.<br><br>FMS data were collected using a sample of 30,000 farms selected to be representative of 500,000 Canadian production enterprises based on major crops: field crops, forage crops, and fruit, vegetable, dairy and nut crops. Strata in the sample were defined by province, and large farms were considered "large farms" because of their national influence on production. |   | 4800               | 1st production       | Intercrop Farm Management FMS | EGF             |          |   | 2017 | 2021 | 8/2/2021 15:52 |

Figure 5: Screenshot of CRDCN dataset database

## Step 1: Metadata crosswalk

- Sounds fancier than it is
- Map fields from my database to the schema from Lunaris
- In some cases these will be dynamic and obvious (ex. “Database name”)
  - Some will be dynamic and non-obvious
- In some cases these will be static and need to be generated (ex. “Rights”)

## Crosswalk examples

- Subjects + Static fields -> Keywords
- Static access data -> Rights
- Permalink (on crdcn.ca) -> URL
- Catalogue ID -> Identifier

## Step 2: Translation code

- Example in the resources (.py and .R) for our translations
- Dynamic fields mapped against the current database
- Static fields added at the end
- If you have EN/FR metadata both can be included in one file
- Write output as .json file



## Step 3: Publish

- Post the .json file somewhere (anywhere) accessible on the web and share this with the Lunaris team
- Updates made to the file will be automatically reflected in Lunaris (“Harvester” checks back on occasion)

# Result

- Restricted dataset is much more discoverable to those seeking information on a given topic
- Updates to metadata including access protocols, dataset info, even new data available are machine readable

# Result

*Lunaris example page*

## Option 2 - Metadata only deposits

- More detailed and of greater use to others
- Create similar metadata profile to previous example
- BUT, can include documentation

# Introducing: Borealis

- National collection of Dataverse instances (one from *most* Canadian universities)
- Your dataverse ingests data, Borealis rolls it up into a national service
- Borealis info is collected by other search tools (including Lunaris).
- *Example*

## Borealis & restricted data

- Can't deposit personally identifiable info into Borealis
- CAN create metadata records including deposit of supporting documents
- With the right supporting documents, Borealis can be a very effective tool for data discovery
  - Anonymized version of dataset
  - Questionnaires
  - Structural files

## Borealis & restricted data

- Metadata only < Metadata w summary docs < Metadata with variable-level information
- Resource document and explainer available

# Why choose Borealis or Lunaris

## Lunaris Pros:

- Dynamic updates to content 'automatic'
- Minimal lift to create a series of entries from existing data

## Borealis Pros:

- Permits a greater level of detail about the resource
- Allows document upload
- Creates a permanent landing page for the resource
- Mints a DOI for the resource which is 'versioned'



# Why choose Borealis or Lunar

## Lunar Cons:

- Limited and inflexible info about data source
- Needs a landing page to point to
- No “permanence” to the record, nothing is created to cite or point to
- System needs some data skill to populate automatically/navigate

## Borealis Cons:

- Process to create/update records more involved
- Many fields to populate (though most are optional)
- Records cannot be deleted (this is also a pro!)
- Accessibility (for upload) limited to consortia members

# When to use Lunaris

- Pre-existing and established set of landing pages
- No additional documentation to upload
- Frequent changes/additions/updates *that don't reflect changes to the underlying data*
- Data provider is not a consortium member

# When to use Borealis

- Want to create a PID for the dataset
- Have additional contextual info to provide (ex. PDF uploads, more thorough metadata)

# Thank you for your attention

- Questions?
- [grant.gibson@crdcn.ca](mailto:grant.gibson@crdcn.ca)